Representation of the Lorentz Group

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Theory

We know that we can define a unitary operator $U(\Lambda)$ that Lorentz transforms a scalar field:

$$U(\Lambda)^{-1}\phi(x)U(\Lambda) = \phi(\Lambda^{-1}x) \longrightarrow U(\Lambda)^{-1}\partial^{\mu}\phi(x)U(\Lambda) = \Lambda^{\mu}{}_{\rho}\,\partial^{\rho}\phi(\Lambda^{-1}x)$$

By analogy to this derivative transformation, we can think of a vector field undergoing such a transform: $U(\Lambda)^{-1}A^{\rho}(x)U(\Lambda) = \Lambda^{\sigma}{}_{\rho}\,A^{\rho}(\Lambda^{-1}x)$

So, why can't we use this for a tensor field?

Basically, we know that vectors, tensors etc. are all just maps b/w spaces. So we are transforming a map.

The tensor field will transform as: $U(\Lambda)^{-1}B^{\mu\nu}(x)U(\Lambda) = \Lambda^{\mu}{}_{\rho}\Lambda^{\nu}{}_{\sigma}B^{\rho\sigma}(\Lambda^{-1}x)$

Both symmetric and antisymmetric nature of a tensor are preserved here, interesting

its trace transforms as:

$$U(\Lambda)^{-1}T(x)U(\Lambda) = U(\Lambda)^{-1}g_{\mu\nu}B^{\mu\nu}(x)U(\Lambda)$$

$$= g_{\mu\nu}U(\Lambda)^{-1}B^{\mu\nu}(x)U(\Lambda)$$

$$= \Lambda^{\sigma}{}_{\mu}\Lambda^{\rho}{}_{\nu}g_{\mu\nu}B^{\rho\sigma}(\Lambda^{-1}x) = g_{\rho\sigma}B^{\rho\sigma}(\Lambda^{-1}x) = T(\Lambda^{-1}x)$$

This tells us that we can potentially breakdown any random tensor into symmetric, antisymmetric and trace components to better understand its LG transform:

$$B^{\mu\nu}(x) = A^{\mu\nu}(x) + S^{\mu\nu}(x) + \tfrac{1}{4}g^{\mu\nu}T(x)$$

Notice how these do not mix. The key takeaway from this example is that, how do we know that the field we are working with is not further reducible? making the analysis easier.

Problem 1: Express $A^{\mu\nu}$, $S^{\mu\nu}$, T in terms of $B^{\mu\nu}(x)$.

$T(x)$ is pretty straightforward: $\underline{T(x) = g_{\mu\nu}B^{\mu\nu}(x)}$

Using this we can see that: $B^{\mu\nu}(x) = A^{\mu\nu}(x) + S^{\mu\nu}(x) + \frac{1}{4}g^{\mu\nu}(g_{\rho\sigma}B^{\rho\sigma}(x))$

$$\longrightarrow B^{\mu\nu}(x) = A^{\mu\nu}(x) + S^{\mu\nu}(x) + \frac{1}{4}B^{\mu\nu}(x)$$

Thus: $A^{\mu\nu}(x) + S^{\mu\nu}(x) = \frac{3}{4}B^{\mu\nu}(x)$

Thus: $\Phi_A(x) + \Phi_A(x) = \frac{3}{4}\Phi_A(x)$

We know, $S^{\mu\nu}(x) = \frac{1}{2}[B^{\mu\nu}(x) + (B^{\mu\nu}(x))^T]$; $A^{\mu\nu} = \frac{1}{2}[B^{\mu\nu}(x) - B^{\mu\nu}(x)^T]$ → Schematically.

For the validity of the decomposition, we need:

$$S^{\mu\nu} = \frac{1}{2}\left\{\frac{3}{4}B^{\mu\nu}(x) + \frac{3}{4}B^{\nu\mu}(x)\right\} \; ; \; A^{\mu\nu} = \frac{1}{2}\left\{\frac{3}{4}B^{\mu\nu}(x) - \frac{3}{4}B^{\nu\mu}(x)\right\}$$

Would work if $S^{\mu\nu}$ was not traceless, since it has a trace you need to do simple index manipulation.

We will now use group representation theory to solve the question posed above. Consider a field with a generic Lorentz Index: $U(\Lambda)^{-1}\Phi_A(x)U(\Lambda) = L_A{}^B(x)\Phi_B(\Lambda^{-1}x)$

We have these matrices $L_A{}^B(\Lambda)$ that need to obey the group decomposition rule:

$L_A{}^B(\Lambda')L_B{}^C(\Lambda) = L_A{}^C(\Lambda'\Lambda)$

We say that these matrices, $L_A{}^B(\Lambda)$ form the representation of a Lorentz group.

We also know that for an infinitesimal transformation:

$$U(1+\delta\omega) = I + \frac{i}{2}\delta\omega_{\mu\nu}\underbrace{M^{\mu\nu}}_{\text{Generators of the LG}} \longrightarrow \underbrace{[M^{\mu\nu}, M^{\rho\sigma}] = i(g^{\mu\rho}M^{\nu\sigma} - (\mu\leftrightarrow\nu)) - (\rho\leftrightarrow\sigma)}_{\text{Commutation relations}}$$

These commutation relations specify the Lie Algebra of the group.

We can also decompose the LG generators to angular mom^n and Boost: $J_i \equiv \frac{1}{2}\epsilon_{ijk}M^{jk}$, $K_i \equiv M^{i0}$

$\longrightarrow [J_i, J_j] = +i\epsilon_{ijk}J_k$, $[J_i, K_j] = +i\epsilon_{ijk}K_k$, $[K_i, K_j] = -i\epsilon_{ijk}J_k$

Notice how these relations are an equivalent set of Lie Algebras.

For an infinitesimal transformation: $L_A{}^B(1+\delta\omega) = \delta_A{}^B + \frac{i}{2}\delta\omega_{\mu\nu}(S^{\mu\nu})_A{}^B$

Putting the above into: $U(\Lambda)^{-1}\Phi_A U(\Lambda) = L_A{}^B(\Lambda)\Phi_B(\Lambda^{-1}x)$

$$\longrightarrow U(1-\delta\omega)\Phi_A U(1+\delta\omega) = \left[\delta_A{}^B + \frac{i}{2}\delta\omega_{\mu\nu}(S^{\mu\nu})_A{}^B\right]\Phi_B(\Lambda^{-1}x)$$

$$\longrightarrow \left\{\Phi_A - \frac{i}{2}\delta\omega_{\mu\nu}M^{\mu\nu}\Phi_A\right\}\left\{1 + \frac{i}{2}\delta\omega_{\mu\nu}M^{\mu\nu}\right\} = \Phi_A(\Lambda^{-1}x) + \frac{i}{2}\delta\omega_{\mu\nu}(S^{\mu\nu})_A{}^B\Phi_B(\Lambda^{-1}x)$$

$$\longrightarrow \Phi_A - \frac{i}{2}\delta\omega_{\mu\nu}M^{\mu\nu}\Phi_A + \frac{i}{2}\delta\omega_{\mu\nu}\Phi_A M^{\mu\nu} = \Phi_A(1-\delta\omega x) + \frac{i}{2}\delta\omega_{\mu\nu}(S^{\mu\nu})_A{}^B\Phi_B(\Lambda^{-1}x)$$

$$\longrightarrow [\Phi_A(x), M^{\mu\nu}] = \mathcal{L}^{\mu\nu}\Phi_A(x) + (S^{\mu\nu})_A{}^B\Phi_B(x)$$

Here, $\mathcal{L}^{\mu\nu} = \frac{1}{i}(x^\mu\partial^\nu - x^\nu\partial^\mu)$. Both $\mathcal{L}^{\mu\nu}$ and $S^{\mu\nu}$ need to obey the same commutation

relations as the generators themselves.

The problem is to find all possible sets of matrices that obey the Lie Algebra of the group.

Also note that while $M^{\mu\nu}$ needs to be hermitian, $(S^{\mu\nu})_A{}^B$ doesn't.
(Angular mom^n only!!)

If we look at the equivalent Lie Algebras, we can see that these are from SU(2) and we already have representations of the group which can be characterized using the angular mom^n number j.

To extend to the whole LA, we define: $M_i \equiv \frac{1}{2}(J_i - iK_i)$; $M_i^\dagger \equiv \frac{1}{2}(J_i + iK_i)$

$[M_i, M_j] = i\epsilon_{ijk} M_k$; $[M_i^\dagger, M_j^\dagger] = i\epsilon_{ijk} M_k^\dagger$; $[M_i, M_j^\dagger] = 0$

We can see that the Lie Algebra of the LA is just two SU(2)s exchanged upon Hermitian conjugation. This allows us to identify the characterization of LA through two indices: $(2n'+1), (2n+1)$ having total no. of components: $(2n'+1)\times(2n+1)$. A few common representations are:

$(1,1) \rightarrow$ Scalar field , $(1,2) \rightarrow$ Right-handed spinor.

$(2,1) \rightarrow$ Left-handed spinor , $(2,2) \rightarrow$ Vector.